

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

B. Tech. (EC)

Sem - IV

University Examination, 2010-11

MA 203: Engineering Mathematics-IV

Date: 09.05.11, Thursday Time: 10:00 a.m to 01:00 p.m

Maximum Marks: 70

Instructions: (1) All questions are compulsory.

(2) Indicate clearly, the option you attempt along with its respective question number.

(3) Figures to right indicate marks

(4) Rough work is done in the last page of main answer book; don't write anything on the question paper.

SECTION-I

Q-1

(a) Find Fourier Sine Transform of the function $f(x) = \frac{e^{-ax}}{x}$. (04)

(b) Find Fourier Cosine Transform of the function $f(x) = e^{-2x} + 4e^{-3x}$ (03)

(c) Find Fourier Transform of the function $f(x) = \begin{cases} 1, & \text{for } |x| < 1 \\ 0, & \text{for } |x| > 1 \end{cases}$. (04)

and hence evaluate integral. $\int_0^{\infty} \frac{\sin s}{s} ds$.

Q-2

(12)

(a) Find an analytic function in terms of z whose real part is $-r^3 \sin 3\theta$.

(b) Show that the transformation $w = \frac{1}{z}$ transforms all circles and straight lines into the circles and straight lines in w -plane, which circles in z -plane becomes straight line in w -plane and which straight lines are transforms into other straight lines.

(c) Define bilinear transformation, Find bilinear transformation which maps the points $(0, 1, \infty)$ onto $(-5, -1, 3)$.

OR

Q-2

(12)

- Show that an analytic function with constant modulus is constant.
- Determine an analytic function $f(z)$ whose real part is given by $u = e^{x^2-y^2} \cos 2y$.
- Sketch the image of rectangle whose vertices are $(0,0)$, $(1,0)$, $(1,2)$, $(0,2)$ by means of linear transformation $w = (1+i)z + 2 - i$.

Q-3

(12)

- Find $L^{-1} \left[\frac{1}{(s+1)(s-2)^2} \right]$ by method of Residues.
- Evaluate: $\int_0^{2+i} (\bar{z})^2 dz$ along the line $y = \frac{x}{2}$.
- State Cauchy integral theorem and evaluate: $\int_{C:|z|=1} \frac{3z^2+z}{z^2-1} dz$.

OR

Q-3

(12)

- Find first four terms of Taylor's series expansion of $f(z) = \frac{z+1}{z^2-7z+12}$ about $z=2$.
- State Cauchy residue theorem and hence evaluate: $\int_{C:|z|=2} \frac{2z-1}{z(z+1)(z-3)} dz$.
- State Cauchy integral formula for derivative and hence evaluate: $\int_{C:|z|=1} \frac{\sin^2 z}{(z-\frac{\pi}{6})^3} dz$.

SECTION-II

Q-4

- Find the Z-transforms of $\left\{ \frac{z^k}{k!} \right\}$. $(k \geq 0)$ (3)
- Solve the difference equation: $y_{k+1} - 2y_{k-1} = 0$, $k \geq 1$, $y(0) = 1$ (4)
- Find the inverse Z-transforms of $\frac{1}{(z-3)(z-2)}$ for $|z| < 2$ (4)

Q-5**(12)**

- (a) Use Langrange's interpolation formula to find the value of y when $x = 5$, if the following values of x and y are given:

x	1	2	3	4	7
y	2	4	8	16	128

- (b) Interpolate by means apposite interpolation formula the population for the year 1936 given the following table:

Year	1901	1911	1921	1931	1941	1951
Population ('000)	12	15	20	27	39	52

- (c) Evaluate $\int_0^4 e^x dx$ by simpson's one-third rule. Compare it with the true value.

OR**Q-5****(12)**

- (a) Evaluate $\int_1^2 \frac{1}{x} dx$ by Trapezoidal rule for $n = 10$.

- (b) Compute $\cosh(0.26)$ using proper interpolation formula

x	0.5	0.6	0.7	0.8
$f(x)$	1.127626	1.185465	1.255169	1.337435

- (c) Evaluate $\int_0^1 e^{-x^2} dx$ by the Gauss integration formula for $n = 3$.

Q-6**(12)**

- (a) Use Newton-Raphson method to solve the transcendental equation $e^x = 5x$.
 (b) Find a real root of the equation $x^3 - 5x - 7 = 0$ using Regula Falsi method.

OR**Q-6****(12)**

- (a) Using Picards method to solve: $\frac{dy}{dx} = x^2 - y$, $y(0) = 1$ for $x = 0.2$.

- (b) Apply the fourth order Runge-Kutta method to solve:

$$\frac{dy}{dx} = x^2 + y^2, \quad y(0) = 1. \text{ Take step size } h = 0.1 \text{ and determine } y(0.1) \text{ and } y(0.2)$$

approximate to correct to four decimal places.